

COINPICKS · RESEARCH & EDUCATION

Bitcoin Technical Analysis Quant Pillar Report

The Tape Itself — Measured, Scored, and Mapped to the Pillar Standard

*Machine-computed from 16 years of daily price data (5,878 days, 2010–2026) through a 7-indicator scoring stack, halving-cycle analogs, and a power-law valuation model. Every number is reproducible from the scripts in the BTC-TA folder. **These numbers are computed, not judged, so they carry no draft/committed status of their own — they enter the master blend directly. Today's overall qualitative score set was left as a draft at delivery this cycle (an adversarial fact-check of the Iran War Risk pillar caught a false claim — a reported "first confirmed fatality" in the tanker campaign was actually roughly the 18th such death, a toll never previously scored — unrelated to this quant pillar) and was COMMITTED BY ALEX on August 23, 2026, after independent review; this pillar's own output is unaffected by that review.***

COINPICKS QUANT PILLAR REPORT · THE QUANTITATIVE SLEEVE OF MY THESIS

Data through August 22, 2026 (BTC \$77,266.84) · Pipeline re-run & verified August 22, 2026 · Not Financial Advice

The Scores — August 22 Model Output

Computed August 22, 2026 (previous computed run August 20, 2026; no cycle ran Aug 21) — same standard as every qualitative pillar. Today's mapping moves every window: 1mo 45→49, 3mo 43→46, 6mo 52→54, 1yr 67→66, 3yr 70→69 — the near-term tape keeps repairing toward flat, with a further one-point pullback at the long end (see *What Changed* for why). Today's overall qualitative score set was left as a draft at delivery (an adversarial fact-check caught a false claim in the Iran War Risk pillar, unrelated to this quant pillar) and was COMMITTED BY ALEX on August 23, 2026, after independent review; this quant pillar is computed fresh every run regardless and carries no draft/committed status of its own.

WINDOW	MODEL P	QUALITY	IMPACT	SIGNAL	CALL
1 month	49	0.75	8	-0.12	Neutral — the tape is nearly flat now (weekly composite +26.4, Slight bullish, 5-of-7) but the current bar doesn't close until tomorrow; the 1mo cycle-analog is still 0-of-3 positive
3 months	46	0.70	8	-0.45	Neutral — up from Mild Bear; still the softest medium window, 1-of-3 analog positive, median -11.9%
6 months	54	0.55	8	+0.35	Neutral — a genuine coin flip, tilting up as 2-of-3 analogs turn positive, median +9.7%
1 year	66	0.60	8	+1.54	Bull — the cycle turn; every prior cycle at this exact month was higher a year later (3-of-3)
3 years	69	0.50	8	+1.52	Bull — value + cycle

Impact: 8 (the tape aggregates every driver, but adds no causal force of its own) · **Formula:** Signal = ((P - 50) ÷ 50) × Quality × Impact · **Honesty cap:** P = 50 + 25 × (window score ÷ 100) — pure TA never claims odds more than 25 points from a coin flip, because its measured out-of-sample edge is modest. Math machine-verified.

THE THREE INPUTS, IN PLAIN ENGLISH

- P** — the odds the tape is *good for Bitcoin* by then. Near-term now even closer to a coin flip (weekly composite +26.4/100, Slight bullish, agreement widened to 5-of-7 as the tape keeps repairing — price is now +20.2% above the 200-week average, though the actual weekly bar doesn't close until tomorrow, Aug 23); long-term still high (at this exact point in prior cycles, the next 12 months were positive 3-for-3, and price sits 44% below its power-law fair value).
- Quality** — how trustworthy the evidence is. Highest near-term (841 weekly bars of history feeding the indicator stack), lowest at 3 years (only 3 completed cycles to learn from, and this cycle is already breaking the old pattern).
- Impact** — 8 out of 10, now the uniform value shared by every pillar; the TA pillar carries an equal 20% weight in the master, same as each qualitative pillar. Price is the thing itself — but a mirror, not an engine.

The Claim

The thesis, and where the numbers differ from the mood

The crowd: -50% from the all-time high just weeks ago, a lower low on June 30, a weekly close under the 200-week average as recently as June 28 — "the cycle is dead, get out." (Price has since closed at \$77,266.84, up **+7.5% versus Aug 20** and +32.0% off the June 30 low, even as the qualitative score set for today was left in draft at delivery after an adversarial fact-check caught a false claim in the Iran War Risk pillar, and was subsequently committed by Alex on

August 23, 2026; the tape moving hard on its own schedule is itself information: this move is coming from price action, not from any of the four qualitative pillars.)

The model: the short-term tape keeps repairing — the weekly composite is now Slight bullish (+9.9→+26.4, agreement widened to 5-of-7) and the quarterly composite firmed further (+18.5→+23.8); the monthly composite is essentially flat (−12.8→−1.9) with agreement narrowing to 2-of-7 as its sign gets muddier near zero. Price is now +20.2% above the 200-week average, but the current weekly bar still hasn't closed (it does tomorrow, Aug 23), and the 1–3 month windows are now Neutral rather than Mild Bear (P 49/46) — this is a further step toward "the correction is over," not yet the confirmation itself. The 6-month window remains a genuine coin flip (P 54). The long read holds its direction but ticked down a further point at each end (1yr 67→66, 3yr 70→69): every prior cycle at this exact month was higher 12 months later, price is still 44% below fair value, and the froth that precedes real crashes remains absent.

The edge: the crowd reads one clock (the tape) — and today, for once, the tape genuinely agrees with the crowd's improving mood. The model still reads three — tape (repairing further, still not confirmed), cycle clock (bottoming window: prior cycles bottomed at months 26–30; we're at month 28.09, just past the average-bottom date), and valuation (44% below the power-law fair line, with price now further above the -1σ "cheap" band than it was two days ago). All three clocks are pointed the same direction on the medium-to-long horizon now — a genuine change from mid-August, when the tape was the lone holdout.

The Facts — The Tape

Computed August 22, 2026 from Bitstamp daily data (BTC \$77,266.84, daily close as of August 22). Current weekly/monthly/quarterly bars are still forming — scores use them as-is. Quantified: on completed bars only, the weekly composite is −29.5, monthly −32.6, quarterly +7.7 — all three unchanged from Aug 20, because Aug 20 and Aug 22 fall inside the same still-forming weekly/monthly/quarterly bars. Nearly all of today's headline improvement below still lives in those forming current bars, not yet in confirmed closed bars.

- **Weekly composite +26.4/100 (Slight bullish), agreement widened to 5-of-7 indicators.** Price is now above the 20-week (\$69,474, +11.2%) and above the **200-week average (\$64,284, +20.2%)** — the headline development this cycle continuing to build (see below) — and has narrowed the gap to the 50-week (\$81,788, −5.5%), a much tighter gap than two days ago; RSI 56.2; MACD line still negative (−3,623) with histogram positive (+1,360) — momentum repair continuing; %B 0.78. Price (\$77,266.84) is also now sitting just 0.6% under the falling weekly resistance trendline (\$77,700) — a near-term test in its own right.
- ★ **Repair signal #1 (a weekly close above the 200-week MA) is now one day from confirmation — but is not yet confirmed.** Price closed Aug 22 at \$77,266.84, +20.2% above the 200-week average (\$64,284) on a daily basis, further above than Aug 20's +11.9%. The actual weekly bar that confirms this signal is still forming (dated 2026-08-23) and closes tomorrow, Sunday; until then this is a strong lean, not a confirmed close. Worth flagging honestly: the completed-bar weekly composite (prior closed bars only, no partial data) is still −29.5, unchanged from Aug 20 — today's improvement, like Aug 20's, is concentrated almost entirely in the current, still-forming bar.
- **Monthly composite −1.9/100 (Neutral), agreement narrowed to 2-of-7** — nearly flat, up from Aug 20's −12.8; the narrower agreement count reflects the composite sitting close to zero, where indicator signs split more evenly. Below the 20-month average (\$88,156, −12.4%), above the 50-month (\$61,025, +26.6%); RSI 50.5; monthly MACD line flipped positive (+100) with histogram −4,858 — still negative but less deeply so; %B 0.34. The completed-bar monthly composite is −32.6, unchanged from Aug 20 — this month's improvement is still concentrated in the still-forming current bar (dated 2026-08-31).
- **Quarterly composite +23.8/100 (Slight bullish), agreement widened to 4-of-7** — firmed further from Aug 20's +18.5. Comfortably above the 20-quarter average (\$58,085, +33.0%) and the 50-quarter (\$28,711, +169.1%); RSI 57.0, MACD line +16,126 with histogram −514 (momentum cooling less than two days ago). Completed-bar quarterly composite is +7.7, unchanged from Aug 20 (current bar dated 2026-09-30, still forming).
- **Levels that matter:** price (\$77,266.84) has broken above the old \$74,434 resistance, which now flips to nearest support (−3.7%, 1 touch), with \$59,930 (−22.4%, 1 touch) and \$57,735 (−25.3%, 1 touch) further below. Overhead

resistance is now \$81,685 (+5.7%, 2x touch), then \$89,164 (+15.4%, 1 touch), then \$98,090 (+26.9%, 2x touch); the falling weekly support trendline sits at \$45,626, the falling resistance trendline at \$77,700 — essentially at today's price.

- ★ **June 30 remains the cycle low: \$58,526 close** — the flush tagged the Fib golden pocket (\$54,257–\$57,802 = 61.8–65% retrace of the \$15,479 → \$126,272 run) and has since built a **+32.0% bounce to \$77,266.84**. **\$58,526 is still the line in the sand** — a decisive break back below it would undo this entire recovery leg.



Weekly tape, scored +26.4/100 (Slight bullish) with agreement widened to 5 of 7 indicators as the tape keeps repairing: price is now above the 20- and 200-week averages and has narrowed the gap to the 50-week sharply, MACD histogram remains positive. Machine-annotated from the same data that feeds the scores.

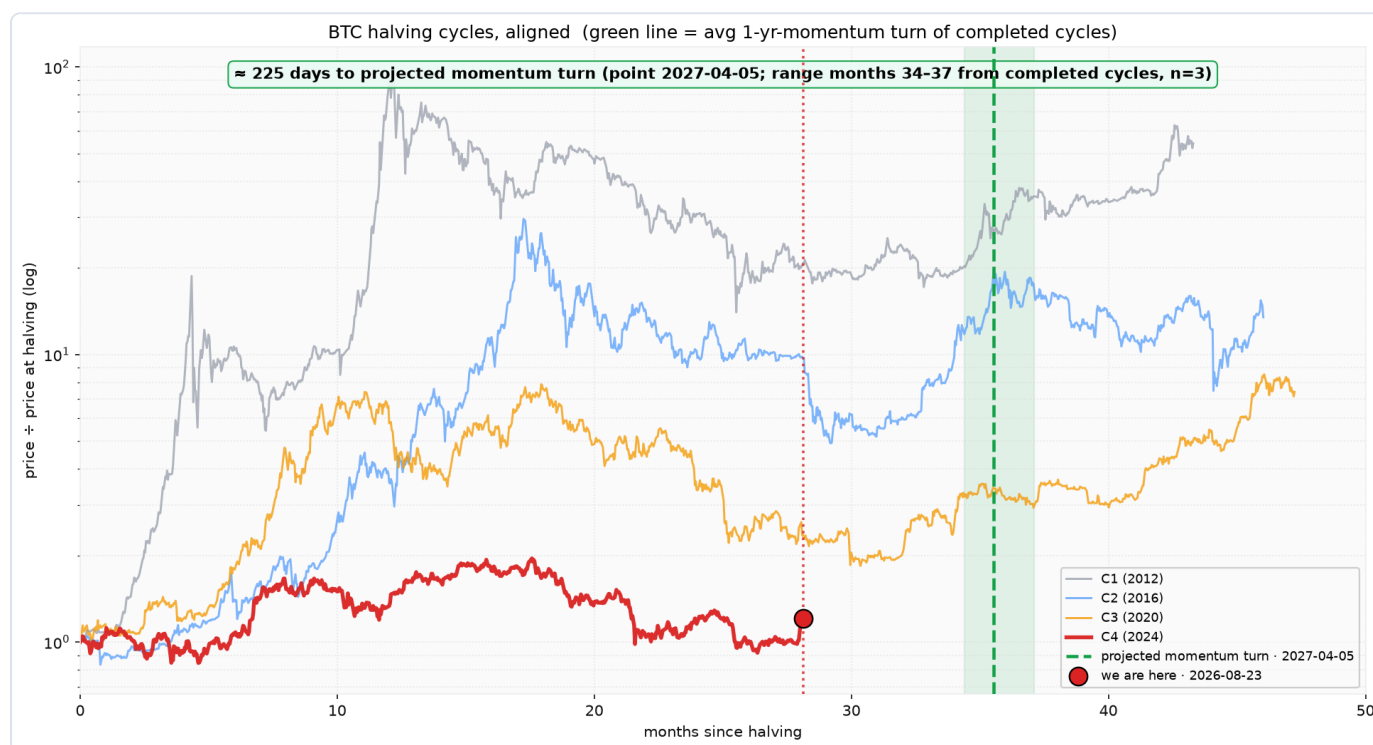
The Facts — The Cycle Clock & The Value Floor

The two slower clocks. Same data, longer lenses.

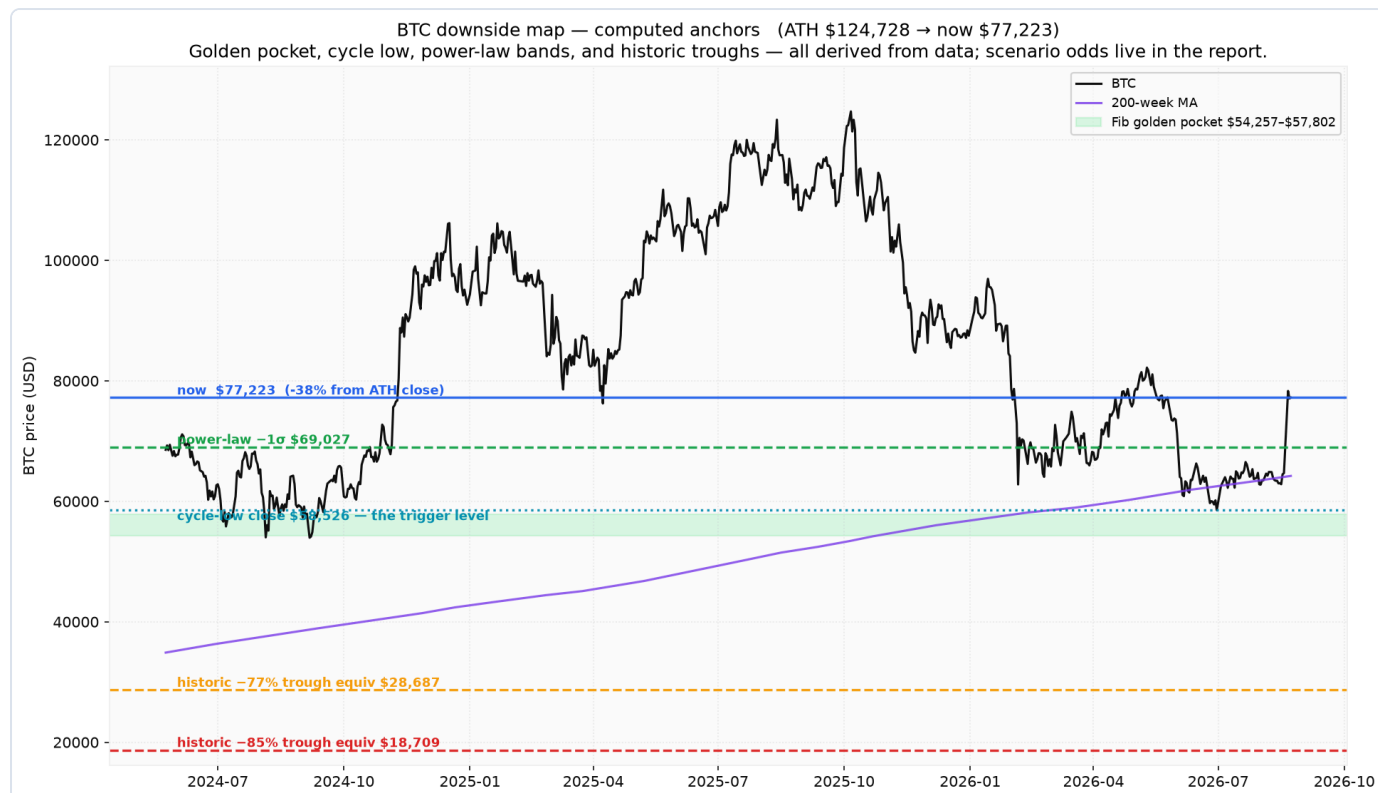
- **The halving clock says "bottoming window."** The three completed cycles topped 12–18 months after their halving (avg 15.8) and bottomed at months 26–30 (avg 28.4). This cycle topped at month 18 (Oct 6, 2025 — \$124,728) and printed its low-so-far at month 26 (June 30, \$58,526, –53% from the top). **We are at month 28.09 — just past the average-bottom date (~Aug 31, 2026), which is now about 9 days out.**
- ★ **The cycle analogs are the spine of the long-window scores — and moved again this cycle, though less dramatically than last check.** Standing at this exact cycle-month in C1/C2/C3: next 1 month positive **0-of-3** (median –7.7%, unchanged agreement, an improved median) · next 3 months positive **1-of-3** (median –11.9%, unchanged agreement, an improved median) · next 6 months positive **2-of-3** (median +9.7%, unchanged agreement, an improved median) · next 12 months positive **3-of-3** (median +37.2%, down slightly from +39.3%) · next 36 months positive **3-of-3** (median +921.2%, down from +968.1%). *Fragility disclosure: this time all five agreement counts held steady as the cycle-month clock advanced from 28.02 to 28.09 — only the medians drifted*

— but that stability is itself just two days of data; treat the analog levels as soft, the direction as the signal. And know what the 36-month number is: from month ~28 that window reaches month ~64 — it contains the NEXT halving's rally.

- **1-year momentum turned positive at months 34–37 in every completed cycle (avg 35.5)** → still projects to ~**April 2027** (80% window Jan–Jul 2027) for this cycle.
- **The power-law floor:** a log-log fit over the full 16-year series puts fair value at **\$138,195** — price is **–44%** versus fair. Fit-window honesty: starting the fit in 2011, 2012, 2013, or 2015 instead moves fair value to \$128,099 / \$133,075 / \$116,618 / \$143,501 respectively (fair value ranges roughly \$117K–\$144K depending on fit window) — the level is soft, but the **"well below fair" verdict holds across every window tested**. Price (\$77,266.84) is now further above the **–1 σ band (\$68,983)** than it was two days ago — further out of the "cheap" zone (the –2 σ band sits at \$34,434, far below). Every halving-era cycle trough was –77% to –85% from ATH (\$18.7K–\$28.7K equivalent today) — the pre-halving 2011 bear went ~–93% (~\$8.7K equivalent), excluded from the range but in the data; this cycle's low-so-far was –53% from the ATH.
- **No froth in the system:** the altcoin-froth index reads $z = -0.87$ — the opposite of the euphoric signature that preceded prior deep crashes. Validated on 16 years of data: high-froth days led to –39% average 12-month drawdowns vs –23% for low-froth days (gap significant, 95% CI excludes zero). *Data lags ~6 weeks (through May 23, unchanged)* — used as a tilt, not a score input.
- ★ **What we rejected:** a 5-feature logistic crash model was tested leave-one-cycle-out and **failed** (AUC 0.44 vs 0.58 for froth alone; 0.70 = useful). It is not in this report's numbers. The scores rest only on what survived out-of-sample testing.



All four halving cycles aligned (log scale, price ÷ price-at-halving). Triangles mark where 1-year momentum turned positive in each prior cycle — months 34–37, projecting to ~April 2027 for this one. The red dot is today: month 28.09, the historical bottoming window.



The downside map, updated after the June 30 flush: the projected \$54K–\$61K bottom zone was hit (\$58,526 close / \$57,735 intraday, inside the golden pocket's doorstep) and bounced. Odds re-weighted accordingly: ~40% the low is in, ~35% one more flush to \$54–58K, ~17% deeper bear, ~8% tail.

The Prize

Illustrative only — not price targets, not advice

From \$77,266.84: the 20-week average (~\$69.5K) is already well behind us — price sits **+11.2%** above it, further along the trend-repair leg; the 50-week average (\$81,788) is the next test, now just **+5.9%** away, and the falling weekly trendline resistance (\$77,700) is closer still, only **+0.6%** away. A retest of the \$124,728 ATH is **+61.4%**; the power-law fair line is **+78.9%**. The median 12-month analog from this cycle-month is **+37.2%** (~\$106.0K by mid-2027); the median 36-month analog is **+921.2%** — **discount that one hard**, it leans on regimes (2017 mania, post-COVID ZIRP) that don't repeat on demand.

The Prize doesn't feed the math — it tells you what the lean is worth acting on. Here it says: the payoff for holding through a now-Neutral (not Mild Bear) near quarter is a full cycle turn — and today's tape read makes that holding period look less lonely still than it did two days ago.

The Other Side

The strongest honest case the model is wrong

- **n = 3.** Every long-window claim rests on three completed cycles. Three coin flips landing heads is not a law of nature.
- **This cycle already broke the pattern.** A -53% drawdown at month 26 vs -77/-85% in prior cycles could mean dampened cycles (bullish read) — or it could mean *the bottom isn't in*: a historic-depth trough from this ATH is \$18.7K–\$28.7K, far below every level in this report.
- **The ETF era changed the machine.** Prior cycles had no spot-ETF flow channel, no CME basis trade at this scale, no corporate treasuries. Analogs may simply not transfer.

- **The power law is a curve fit, not a promise.** R^2 0.96 in-sample says nothing about a structural break — a fit like this "said cheap" through most of 2022's decline too. It bounds scenarios; it doesn't time anything.
- **The tape's edge is small by construction.** The honesty cap exists because measured predictive validity is modest (froth AUC 0.58). If you removed the cap, this report would be louder — and less true.

The Triggers

Price and dated events — some prove the read right, some prove it wrong

WHEN / LEVEL	EVENT	MEANS
\$58,526	Daily close below the June 30 low	Flush scenario live — golden pocket \$54.3–57.8K is the target; re-rate 1/3-month down
~\$64,284	Weekly close above the 200-week MA	One day from confirmation — the daily close (Aug 22) already sits +20.2% above this level; the actual weekly bar (dated 2026-08-23) closes tomorrow, Sunday — the real remaining test, now imminent
~\$69,474	Weekly close above the 20-week MA	Daily close is already clear (+11.2%); a confirmed Sunday weekly close above it would lock in trend repair — re-rate 1/3-month up if it holds
~Aug 31, 2026	Historical average-bottom month (28.4) passes	Arriving in ~9 days with price already well off the June 30 low (+32.0%); a higher low + improving monthly momentum after this date = basing confirmed
Jan–Jul 2027	Projected 1-yr momentum turn window (point: ~Apr 2027)	Arriving without new lows = cycle thesis on track; re-rate 1-year up
\$50,000	Weekly close below	–60% from ATH; the dampened-cycle story is failing — see the flip line

The flip line: a decisive weekly close below **\$50,000** breaks the base case and the golden pocket both — treat the dampened-cycle thesis as broken and put the historic trough zone (\$18.7K–\$28.7K) back on the table. Above it, a bearish quarter is the plan, not a surprise.

My Decision

QUALITATIVE SET: COMMITTED BY ALEX AUGUST 23, 2026 · THIS PILLAR: COMPUTED AUGUST 22, 2026 (NO DRAFT/COMMITTED STATUS OF ITS OWN)

Computed P: 49 / 46 / 54 / 66 / 69 · Quality: 0.75 / 0.70 / 0.55 / 0.60 / 0.50 · Impact: 8

This is the quantitative sleeve of the thesis — the same five windows, same formula, same table as every qualitative pillar, but computed instead of judged. The near-term read has improved further today: the weekly and quarterly composites both firmed into Slight bullish territory and price extended its move above the 200-week average on a daily basis, though the weekly bar itself still hasn't closed and the 1–3 month windows are now Neutral (up from Mild Bear). The 6-month window remains a genuine coin flip (P 54). The long read ticked down a further point at both ends (1yr 67→66, 3yr 70→69) even as everything short-term improved — a reminder that these windows partly price in how much of the value/analog gap has already closed, so a rally can pull long-window P down a touch even while the near-term score rises.

These numbers enter the master blend exactly as computed — this pillar has no override/re-stamp step of its own. Unlike the four qualitative pillars, this pillar carries no draft/committed status of its own: today's overall qualitative score set was left as a draft at delivery — not for any reason in this report, but because an adversarial fact-check of the Iran War Risk pillar caught a false claim (a reported "first confirmed fatality" in the tanker campaign that was actually roughly the 18th such death, a toll never previously scored) that the original research pass did not catch. Per house discipline, verification surfacing an inconsistency it can't fully resolve sends the whole score set to draft rather than partial self-approval. Alex independently reviewed the math and research trail and **committed the set on August 23, 2026**. This pillar's own computed output is unaffected by that review and enters the master blend the same way it does every day, draft or committed.

What Changed Since Last Check

Pipeline re-run August 22, 2026 (previous computed run August 20, 2026; no cycle ran Aug 21).

- **Full-weekly-refresh re-pull, same day: \$77,169.40 (this morning's catch-up-cycle delivery) → \$77,266.84 (this refresh), +0.13%, essentially flat.** Same computed P/Quality/Signal table (49/46/54/66/69) and the same composite calls; indicator, level, and derived-percentage figures below are refreshed to this exact re-pull.
- ★ **Price continued higher: \$71,882.92 (Aug 20) → \$77,266.84 (Aug 22), +7.5%.** June 30's \$58,526 remains the cycle low; price is now +32.0% off it.
- ★ **Repair signal #1 extended: price moved from +11.9% to +20.2% above the 200-week average.** Today's close (\$77,266.84) sits +20.2% above the updated 200-week MA (\$64,284) on a daily basis. The actual weekly bar (dated 2026-08-23) still hasn't closed — it does tomorrow, Sunday — so this remains one day from confirmation, not yet confirmed.
- **Composites firmed further on the current (forming) bar; completed-bar figures held flat, as expected since Aug 20 and Aug 22 sit in the same forming weekly/monthly/quarterly bars.** Weekly +9.9 → +26.4 (agreement widened 4-of-7 → 5-of-7, now Slight bullish), monthly −12.8 → −1.9 (agreement narrowed 3-of-7 → 2-of-7, essentially flat), quarterly +18.5 → +23.8 (agreement widened 3-of-7 → 4-of-7, now Slight bullish). On completed bars only, all three held exactly flat: weekly −29.5, monthly −32.6, quarterly +7.7 — confirming that today's improvement, like Aug 20's, is still concentrated in the still-forming current bars.
- **Model P: 45/43/52/67/70 → 49/46/54/66/69.** The three near-term windows continued moving up (1mo +4, 3mo +3, 6mo +2) on the further tape repair. The two long windows each ticked DOWN a further point (1yr 67→66, 3yr 70→69) even as everything else improved — the same honest artifact as last cycle: price closing more of the value/analog gap (price moved from roughly −48% to −44% vs. power-law fair value, and further above the -1σ "cheap" band) leaves slightly less "already-cheap" edge to price in at the long end.

- **Cycle analogs drifted less this time:** all five agreement counts (0/3, 1/3, 2/3, 3/3, 3/3) held steady as the cycle-month clock advanced from 28.02 to 28.09; medians moved modestly (1mo -12.7%→-7.7%, 3mo -21.7%→-11.9%, 6mo +0.3%→+9.7%, 1yr +39.3%→+37.2%, 3yr +968.1%→+921.2%) — flagged per this report's standing n=3 fragility disclosure, not treated as new structural information.
- **Data refreshed:** Bitstamp daily through August 22, 2026 (5,878 days on the master series); Coin Metrics altcoin set still through May 23 (community data lags ~6 weeks — froth is a tilt, not a score input).
- **Context:** today's overall qualitative score set was left as a draft at delivery — an adversarial fact-check of the Iran War Risk pillar caught a false claim (a reported "first confirmed fatality" in the tanker campaign, actually roughly the 18th such death, never previously scored) that the original research pass missed, unrelated to this quant pillar; the set was subsequently **committed by Alex on August 23, 2026**. This pillar's own computed output is independent of that process and enters the master blend the same way regardless of the qualitative set's status.

Method & Sources

Everything is reproducible — the scripts and data live in the BTC-TA folder. Run order below.

- **Price data:** Bitstamp OHLC API (daily, 2011–2026, no key) via [fetch_btc_history.py](#) · Coin Metrics community reference price (2010–2011 close-only) via [build_master_history.py](#) — 5,835 days total, each row tagged by source.
- **Indicator stack:** [indicators.py](#) — SMA 20/50/200, EMA 12/26, RSI-14 (Wilder), MACD 12/26/9, Bollinger 20/2, OBV — on weekly (W-SUN), monthly, quarterly bars. [levels.py](#) — swing-cluster S/R + least-squares trendlines.
- **Scoring:** [score.py](#) — each indicator → conviction in [-100, +100] (bounded inputs mapped linearly, unbounded squashed with tanh); composite = fixed weights (MA .20, RSI .15, MACD .15, BB .15, S/R .15, OBV .10, TL .10); agreement = indicators sharing the composite's sign.
- **Cycle & value:** [cycle_analysis.py](#) — halving-aligned tops/bottoms/momentum turns, projections from C1–C3 · [downside.py](#) / [downside_map.py](#) — power-law bands (R^2 0.96) + historical troughs + 6-method bottom-zone synthesis · [test_altcoin_froth.py](#) — froth quartiles vs forward drawdowns, block-bootstrap CI.
- **The mapping (new):** [pillar_mapping.py](#) — window scores from tape composites + cycle analogs + power-law value; $P = 50 + 25 \times (S/100)$; Quality rubric per window; writes data/pillar_scores.json for the master blend.
- **Run order:** fetch_btc_history → build_master_history → fetch_altcoin_data → indicators → levels → score → charts → cycle_analysis → downside → downside_map → pillar_mapping.
- **Honesty notes:** current weekly/monthly/quarterly bars are partial as of Aug 22 (completed-bar composites: weekly -29.5 / monthly -32.6 / quarterly +7.7 — all three unchanged from Aug 20, because Aug 20 and Aug 22 fall inside the same still-forming bars; nearly all of today's current-bar improvement is still "forming bar" noise until bars close) · the multi-signal crash model failed out-of-sample (AUC 0.44) and is excluded · the AUC figures are the pipeline's own attestation · froth data lags ~6 weeks · n=3 on all cycle claims, and analog windows can cross into the next cycle — the 6-month analog remains fragile on pure alignment drift, printed in the Facts · the 2024 halving is dated Apr 19 (US ET; block 840,000 hit Apr 20 00:09 UTC — 0.03 cycle-month effect) · S/R levels are algorithmic swing-clusters with 1–2 touches each · the bottom-odds split (40/35/17/8) is a judgment synthesis, not a model output.